Together First, Apart Later? When Optimizers Start to Learn Different Functions

Initial checkpoint, MIT 6.7910, Fall 2026

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Summary. We train one transformer from one initialization with AdamW and with Muon (Jordan et al., 2024), on the same sequence of mini-batches. (1) At equal training loss, AdamW and Muon predict as similarly as two runs of one optimizer on different data orders; the loss barrier between their final weights exceeds the data-order ones in only one of our three settings. (2) During training, the singular values of their weight and update matrices differ far more than between data orders (Muon's nearly equal, AdamW's dominated by a few large ones), while their prediction statistics do not. (3) A run switched to the other optimizer mid-training ends, in its predictions, no further from either optimizer's own run than two data orders are, while its weights keep part of the first optimizer's singular-value structure.

Setup. The main setting, TinyStories-10M, trains a 14M-parameter nanoGPT (Karpathy, 2022) (10.6M without embeddings and head) on TinyStories (Eldan and Li, 2023) for 10,000 steps of 128 sequences × 256 tokens with a cosine learning-rate decay; two more settings, a 30M model on TinyStories (TinyStories-30M) and a character-level model on Shakespeare, serve as checks (Appendix A). Muon orthogonalizes the update of each weight matrix in the blocks (sets its singular values to about 1); a separate AdamW trains the embeddings, output head and LayerNorm gains; each optimizer uses the best of five learning rates tried. All runs start from the same initial weights. A *data order* is a fixed choice of mini-batch for every step; each optimizer is run on five. The distance between two runs is the mean Jensen–Shannon divergence (JS, in bits) of their next-token predictions on 2,048 fixed held-out positions. We compare two runs at the checkpoints where they reach the same *training loss* (on 64 fixed training sequences), so that a faster optimizer is not counted as different, and divide their distance by a *floor*: the mean distance between two data orders of one optimizer (for two optimizers, the larger of their floors). A ratio near 1 thus means "as different as two data orders". For any other metric, the *score* of two optimizers is likewise their gap on the same data order over the gap between two data orders of one optimizer, signed (positive when the first is higher) and taken as the median over training losses 1.66–3. The *loss barrier* between two weight vectors is the largest amount, over 11 evenly spaced points on the straight line between them, by which the training loss exceeds the straight line joining its two end values (Frankle et al., 2020); with a near-zero barrier they are *linearly connected*. Losses and barriers are in nats. In a *swap*, a run of one optimizer (the *parent*, data order 0) is switched to the other at a *fork step* and trained to step 10,000 on the parent's batches; the new optimizer starts from a *warm state*, its momentum statistics gathered alongside the parent at learning rate 0. Its *own run* is the run trained with it from the start. In total we trained 141 runs using 33 GPU-hours (Appendix D).

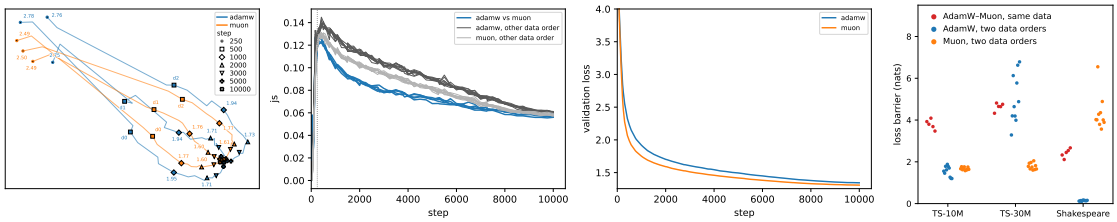


Figure 1: AdamW (blue) and Muon (orange); first three panels on TinyStories-10M. First: checkpoints from step 250 on (data orders d0–d2, at least 0.28 apart along each run), placed in 2-D by densMAP from $\sqrt{\text{JS}}$; numbers: training loss at steps 250, 1000, 2000. The map shows which checkpoints are nearest, not how far apart they are: the six endpoints, drawn almost on top of each other, are JS 0.056–0.063 apart, about as far as each run moves between steps 5000 and 10,000. Second: JS between AdamW and Muon on the same data order at the same step (blue, five data orders) and between two data orders of one optimizer (grey); dotted: step 250. Third: validation loss, mean and range over five data orders. Right: loss barrier between two final weight vectors (how far the training loss rises along the straight path between them above the straight line joining the two end losses), one point per pair: AdamW and Muon on the same data order (red, 5 pairs), two data orders of AdamW (blue) or of Muon (orange; 10 pairs each).

Step 1: how the trajectories differ (Fig. 1). After the learning-rate warmup, AdamW and Muon runs on the same data order stay close: each such pair are nearest neighbours in the densMAP plot (Fig. 1, left), and at every step their JS is smaller than between two data orders of one optimizer (second panel). Muon learns faster (third panel): by step 7500 it reaches AdamW's final validation loss, 1.344 (means over five data orders). Compared at equal training loss, the AdamW–Muon distance is about equal to the floor: at training losses below 4.85 it is 0.95–1.38 times the floor (median 1.02), and 0.99 times at the lowest loss all runs reach (1.02 on each of TinyStories-30M and Shakespeare). So, judged by their predictions, AdamW and Muon end as similar to each other as two runs of one optimizer on different data orders.

Their final weights, however, are further apart than two data orders' are, at least on TinyStories-10M (Fig. 1, right): the loss barrier is 3.5–4.1 between AdamW and Muon, against 1.2–1.9 between two data orders of AdamW and 1.6–1.8 between two of Muon. On TinyStories-30M and Shakespeare, the mean AdamW–Muon barrier lies between the two optimizers' mean data-order barriers (4.64 between 1.78 and 5.05; 2.42 between 0.14 and 4.34).

Step 2: what differs along the way. We measure 31 properties of each model at 34 points during training, from the first steps to the final model (all listed in Appendix B). They fall into five groups: the weights themselves, the change made by a single update step, the curvature of the loss, the internal activations, and the predictions. As with the predictions, we compare each at equal training loss and give it a score, with three data orders per optimizer (Fig. 2 and Table 4, which also include SGD with momentum for reference).

The metrics of the weight matrices' singular values (spectral norm, stable and effective rank, singular-value spread) differ between AdamW and Muon 74–263 times as much as between data orders. The prediction metrics (entropy, confidence, calibration, accuracy, loss on frequent and rare targets) differ at most 1.1 times as much, and how far the weights move barely differs either: 2.18 and 2.155 times the initial weight norm $\|\theta_0\|$. Below, values are at training loss 1.81 and scores are in parentheses. Muon's weight matrices have more evenly sized singular values: their stable rank $\|W\|_F^2/\|W\|_2^2$ is 86 against AdamW's 33 (score −147), and their largest-to-median singular-value ratio $\sigma_1/\sigma_{\text{median}}$ is 2.7 against 8.0 (105). AdamW's model uses fewer MLP units, has larger activations and sharper attention, and sits where the loss is less curved: 12% of MLP units are active against 22% (−62), the residual-stream RMS is 4.7 against 3.4 (28), attention entropy is 2.5 against 2.8 nats (−18), and the Hessian trace $\operatorname{tr} H$ is 1400 against 2770 (−23; top eigenvalue 6.2 against 14, −10). Muon's one-step updates have a far higher stable rank (220 against 17, −328), by construction. SGD moves differently from both: it ends $0.49\|\theta_0\|$ from the initialization (scores 50 against AdamW and 58 against Muon), its steps are 0.06% of the weight norm $\|\theta\|$ against 0.66% (67, 132), they align better with the current batch's gradient (cosine 0.45 against 0.24 and 0.22; −17, −18), and its loss is more curved ($\operatorname{tr} H$ 25,500; −22, −20).

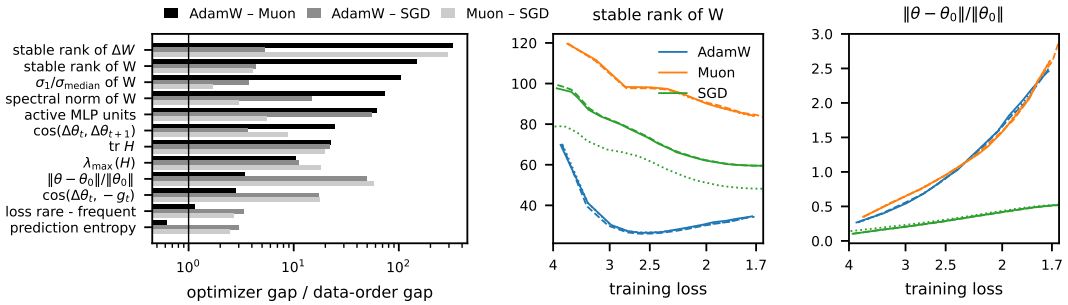


Figure 2: TinyStories-10M, three data orders per optimizer. Left: absolute score of each pair of optimizers on selected metrics (log scale; all metrics, with signs, in Table 4). Δ: change over one step; g_t: gradient on the step's batch; H: Hessian on 32 training sequences. Middle and right: stable rank of W and distance from initialization $\|\theta - \theta_0\|/\|\theta_0\|$ against training loss; colour: optimizer, line style: data order.

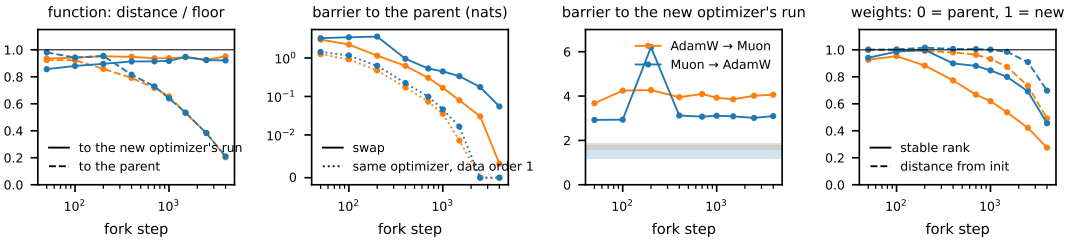


Figure 3: Swaps on TinyStories-10M, data order 0, against the fork step; each panel shows a property of the swapped run's final weights. Left: JS from the swapped run to the new optimizer's own run and to the parent, each divided by the floor of that run's optimizer. Second: loss barrier to the parent, for the swapped run and, as a control, for the parent's own optimizer continued from the same step on data order 1. Third: loss barrier to the new optimizer's own run; bands: barriers between two data orders of that optimizer. Right: stable rank and distance from initialization, rescaled so that 0 is the parent's final value and 1 the final value of the new optimizer's own run.

Step 3: swaps (Fig. 3). We swap at nine fork steps from 50 to 4000, in both directions (Appendix C). Every swapped run ends closer to the new optimizer's own run than two data orders are (0.86–0.95 of the floor), and also closer to the parent, more so the later the swap (0.92–0.98 of the floor at step 50, 0.20–0.21 at step 4000): its predictions are as similar to either optimizer's run as two data orders are. Its weights, in contrast, do not end close to both runs. A swapped run is linearly connected to its parent (barrier below 0.05) from fork step 2500 when leaving AdamW and at no fork step when leaving Muon (0.055 at step 4000), whereas a run switched only to data order 1 is from step 1000. No swapped run ends with a barrier to the new optimizer's own run as small as the barriers between that optimizer's data orders (third panel). Switching at step 4000 moves the final stable rank only 0.27 (AdamW to Muon) and 0.46 (Muon to AdamW) of the way to the new optimizer's value, and the distance from initialization 0.49 and 0.70.

Next. (i) Steps 2 and 3 on TinyStories-30M and Shakespeare; (ii) do the weight spectra explain the AdamW–Muon barriers? (iii) talk (Dec 1–4); paper, code, Colab (Dec 5–11). https://github.com/qqgjyx/MIT-6.7910

References

Eldan, R. and Y. Li (2023). *TinyStories: How Small Can Language Models Be and Still Speak Coherent English?*

Frankle, J., G. K. Dziugaite, D. M. Roy, and M. Carbin (2020). "Linear Mode Connectivity and the Lottery Ticket Hypothesis". In: *ICML*.

Jordan, K., Y. Jin, V. Boza, et al. (2024). *Muon: An Optimizer for Hidden Layers in Neural Networks*. Blog post. URL: https://kellerjordan.github.io/posts/muon/.

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A Hyperparameters

Table 1: Settings shared by the AdamW and Muon runs on TinyStories-10M. Values from the repository's configuration files and code.

Runs	AdamW and Muon × data orders d0–d4 (10)
Model	nanoGPT, pre-LayerNorm blocks
Layers, heads, width	6, 6, 384
MLP	GELU, width 1536
Biases, dropout	none, none
Output head	untied
Attention projections	separate query, key, value
Context	256 tokens
Parameters	13.9M (10.6M without token and position embeddings and head)
Data	TinyStories (GPT-4 version)
Tokenizer	byte-level BPE, 4096 tokens, trained on the training split
Batch	128 windows × 256 tokens
Window offsets	from the training split; generator seeded with (data order, step), same for every optimizer
Steps	10,000
Tokens	328M (about 0.6 epochs)
Learning rate	$\eta\,\varphi(t)$ at step t, every parameter group
Warmup	linear, $\varphi(t) = (t+1)/200$ for $t < 200$
Decay	cosine to $\varphi = 0.1$ at step 10,000
Gradient clipping	none
Initialization seed	0, all runs
Training precision	bfloat16 autocast
Prediction, Hessian precision	float32, no autocast (TF32 matrix products allowed)
Logged predictions	next-token distributions: 128 fixed validation sequences × 16 evenly spaced positions (2,048), as float16 log-probabilities
Training loss	64 fixed training sequences, all positions; used to match runs at equal loss
Both logged at	every 250th step + 30 log-spaced steps in $[1, 10{,}000]$
Hessian	top eigenvalue, 20 Lanczos iterations, 32 fixed training sequences; AdamW also of $P^{-1/2}HP^{-1/2}$, with P AdamW's diagonal preconditioner
Measured at	every 500th step + 30 log-spaced steps
Loss barrier	64 fixed training sequences, 11 evenly spaced points of the segment between the final weights

Table 2: Optimizer-specific settings of the TinyStories-10M runs. n_{out}, n_{in}: output and input size of the matrix.

	AdamW	Muon
η	$2 \cdot 10^{-3}$	0.01
η choice	best final validation loss of a 5-rate sweep on d0	as AdamW
Update	bias-corrected moments	Nesterov momentum; 5 quintic Newton–Schulz steps in bfloat16
Momentum	$\beta = (0.9, 0.95)$	$\beta = 0.95$
ϵ	10^{-8}	–
Update scale	–	$\sqrt{\max(1, n_{\text{out}}/n_{\text{in}})}$
Weight decay	0.1, decoupled; none on LayerNorm gains	none
Parameters	all	query, key, value, attention-output and both MLP matrices of every block
Aux. AdamW	–	embeddings, head, LayerNorm gains
Aux. η, β, ϵ	–	$2 \cdot 10^{-3}$, $(0.9, 0.95)$, 10^{-8}
Aux. weight decay	–	none

Table 3: Shakespeare-char and TinyStories-30M: rows that differ from TinyStories-10M; all else as in Tables 1 and 2. "Aux.": Muon's auxiliary AdamW.

	Shakespeare-char	TinyStories-30M
Runs	AdamW, Muon × d0–d4 (10)	AdamW, Muon × d0–d4 (10)
Data	Shakespeare, nanoGPT's 90/10 split	TinyStories
Vocabulary	65 characters	4096 (BPE)
Layers, heads, width	6, 6, 384	8, 8, 512
Parameters (without embeddings, head)	10.8M (10.6M)	29.5M (25.2M)
Batch	64×256	128×256
Steps	1,500	20,000
Tokens	24.6M (about 24 passes over the 1M training characters)	655M (about 1.2 epochs)
Warmup steps	100	400
Cosine decay to step	1,500	20,000
η_{AdamW}	$3 \cdot 10^{-4}$	$2 \cdot 10^{-3}$
η_{Muon} (aux.)	0.04 (10^{-3})	0.01 ($1.5 \cdot 10^{-3}$)
η chosen by	minimum validation loss over training (runs overfit)	best final validation loss
AdamW β	$(0.9, 0.99)$, as nanoGPT	$(0.9, 0.95)$
Predictions	all 256 positions of 64 validation sequences (16,384)	16 positions of 128 sequences (2,048)
Predictions at	every 50th + 40 log-spaced	every 500th + 30 log-spaced

B Metrics

The Step-2 runs save checkpoints every 1000 steps, at 12 log-spaced steps, and at three consecutive steps $t, t+1, t+2$ for 24 values of t from 25 to 9998 (to measure one-step updates); we measure the metrics at 34 of these per run, and the Step-3 swaps only at their final weights. *Weights*: for all parameters together, $\|\theta\|_2$, $\|\theta - \theta_0\|/\|\theta_0\|$ and $\|\theta\|_\infty/\|\theta\|_2$; and, averaged over the 36 weight matrices W of the blocks, the stable rank $\|W\|_F^2/\|W\|_2^2$, the effective rank $\exp H(\sigma/\sum\sigma)$ with H the entropy of the normalized singular values σ, the spectral norm, and $\sigma_1/\sigma_{\text{median}}$. *Updates* $\Delta\theta_t = \theta_{t+1} - \theta_t$: $\|\Delta\theta_t\|/\|\theta_t\|$, its cosine with $\Delta\theta_{t+1}$ and with the negative gradient $-g_t$ on step t's batch, and the stable and effective rank of its block matrices ΔW. *Curvature* of the training loss on 32 fixed training sequences, with Hessian H: $\lambda_{\max}(H)$ (20 Lanczos iterations, logged during training), $\operatorname{tr} H$ (10 Rademacher probes), $\operatorname{tr} H/\lambda_{\max}$, and the batch sharpness $\mathbb{E}_B[g_B^\top H_B g_B]/\mathbb{E}_B\|g_B\|^2$, where g_B and H_B are the gradient and Hessian on a mini-batch B. *Predictions*, on the 2,048 held-out positions: entropy, mean top-1 probability, calibration error (15 bins), top-1 accuracy, loss on the targets in the top and the bottom third of training-set frequency and their difference, JS to the unigram and to the bigram model of the training split, and validation minus training loss. *Activations*, on 32 held-out sequences: attention entropy and attention on the first token (means over layers, heads and positions), fraction of MLP pre-activations above 0, RMS of the residual stream after the last block, and effective rank of the centred final features.

Table 4: All metrics of Step 2 on TinyStories-10M (Fig. 2). Columns AdamW, Muon and SGD give each metric's value at training loss 1.81, as the mean ± half the range over data orders 0–2. Columns A–M, A–S and M–S give each pair's score: their gap on the same data order (mean over data orders) divided by the larger of the two optimizers' mean gaps between data orders, median over training losses 1.66–3; it is positive when the first optimizer's value is higher. Bold: the five largest in absolute value per column. SGD uses heavy-ball momentum 0.9 and learning rate 0.06, the only rate tried between 0.03 and 0.12 that did not diverge on any of the three data orders; everything else is as in Table 1.

	AdamW	Muon	SGD	A–M	A–S	M–S
Weights						
weight norm $\|\theta\|_2$	224 ± 2	224.3 ± 0.5	109.5 ± 0.1	−0.3	**51**	**63**
distance from init. $/\|\theta_0\|$	2.18 ± 0.03	2.155 ± 0.006	0.486 ± 0.006	−1.3	**50**	**58**
largest entry $/\|\theta\|_2$	0.00704 ± 0.00006	0.00645 ± 0.00002	0.0395 ± 0.0010	9.1	−23	−23
stable rank of W	33.0 ± 0.3	86.2 ± 0.3	56.2 ± 5.6	**−147**	−4.4	4.1
effective rank of W	272.2 ± 0.1	324.5 ± 0.1	321.0 ± 0.9	**−263**	−25	2.2
spectral norm of W	5.39 ± 0.07	2.949 ± 0.009	1.47 ± 0.11	74	15	3.0
$\sigma_1/\sigma_{\text{median}}$ of W	8.01 ± 0.05	2.685 ± 0.005	4.37 ± 0.55	**105**	3.7	−1.7
Updates (one step)						
step size $\|\Delta\theta\|/\|\theta\|$	0.00656 ± 0.00008	0.00653 ± 0.00003	0.000624 ± 0.000047	−3.1	**67**	**132**
stable rank of ΔW	16.7 ± 0.3	219.6 ± 0.3	8.37 ± 0.48	**−328**	5.3	**292**
effective rank of ΔW	228 ± 1	369.4 ± 0.2	205 ± 2	**−135**	3.7	46
cosine of consecutive steps	0.883 ± 0.002	0.738 ± 0.003	0.857 ± 0.004	25	3.6	−8.8
cosine of step and $-g$	0.244 ± 0.004	0.217 ± 0.007	0.447 ± 0.010	2.8	−17	−18
Curvature						
top Hessian eigenvalue	6.18 ± 0.10	14.3 ± 0.2	50.2 ± 0.6	−10	−11	−18
batch sharpness	0.974 ± 0.254	3.13 ± 0.37	6.38 ± 0.50	−2.7	−5.6	−2.8
Hessian trace	1400 ± 40	2770 ± 20	25500 ± 400	−23	−22	−20
trace / top eigenvalue	224 ± 6	196 ± 7	507 ± 9	−0.5	−7.4	−8.7
Activations						
attention entropy (nats)	2.51 ± 0.03	2.79 ± 0.02	3.84 ± 0.01	−18	−26	−17
attention on first token	0.0175 ± 0.0011	0.0198 ± 0.0005	0.0182 ± 0.0007	−1.3	−1.0	1.2
active MLP units	0.1199 ± 0.0009	0.222 ± 0.001	0.262 ± 0.002	−62	**−56**	−5.5
residual-stream RMS	4.73 ± 0.05	3.38 ± 0.02	0.244 ± 0.003	28	**78**	**81**
effective rank of features	312 ± 2	319.6 ± 0.2	297 ± 2	−13	−2.7	4.4
Predictions						
prediction entropy (nats)	1.887 ± 0.008	1.878 ± 0.010	1.943 ± 0.005	−0.2	−3.0	−2.5
mean top-1 probability	0.561 ± 0.003	0.563 ± 0.001	0.5581 ± 0.0008	−0.4	−0.3	0.3
calibration error	0.0325 ± 0.0017	0.0305 ± 0.0046	0.0297 ± 0.0028	0.1	0.5	0.4
top-1 accuracy	0.546 ± 0.005	0.544 ± 0.005	0.554 ± 0.003	−1.0	−2.2	−1.2
loss on frequent targets	0.896 ± 0.004	0.892 ± 0.008	0.851 ± 0.012	0.5	2.5	1.9
loss on rare targets	3.10 ± 0.02	3.102 ± 0.009	3.11 ± 0.04	−0.8	−2.7	−2.1
rare minus frequent loss	2.21 ± 0.02	2.21 ± 0.01	2.26 ± 0.03	−1.1	−3.3	−2.7
JS to unigram model	0.8004 ± 0.0007	0.802 ± 0.001	0.804 ± 0.001	0.8	0.4	−0.7
JS to bigram model	0.4691 ± 0.0010	0.4708 ± 0.0008	0.4622 ± 0.0009	−0.1	2.4	2.3
validation minus training loss	−0.0217 ± 0.0092	−0.0197 ± 0.0062	−0.0224 ± 0.0074	−0.2	0.1	0.3

C Swaps

Table 5: Swap forks on TinyStories-10M. A: parent's optimizer; B: optimizer after the swap; s: fork step.

Parents	the d0 runs of AdamW and Muon (Table 1)
Shadow state	the other optimizer, stepped on the same mini-batch gradients with learning rate 0: momenta, second moments and step counts evolve as if training; parameters follow A alone
Fork steps s	50, 100, 200, 400, 700, 1000, 1500, 2500, 4000
Saved at s	parameters; states of both optimizers
Swap $A \to B$ loads	parameters at s; B's shadow state
Hyperparameters	B's (Table 2)
Steps	s to 10,000, on the parent's schedule and batches (data order 0)
Swap forks	2 directions × 9 steps = 18
Same-batch control	the parent itself (A continued with the same batches)
Data-order control	A continued from s with its own state, data order 1: 18 forks

D Compute

All runs used one NVIDIA GPU each on the MIT ORCD cluster. GPU-hours are those Slurm allocated to each suite (a group of runs launched together); where runs from several suites shared a GPU, its hours were split among them in proportion to training time. We did not count the measurements made after training (Muon's spectral-norm sharpness, loss barriers, distances and the metrics of Appendix B), which took a few GPU-hours.

Table 6: Runs described in the checkpoint and their compute.

	Runs	GPU-hours
Learning-rate sweeps, AdamW and Muon (3 settings)	30	7.1
Runs from the initialization, AdamW and Muon (3 settings, 5 data orders)	30	11.6
Swaps and data-order forks (Table 5)	36	7.1
Step 2: SGD learning-rate search, SGD runs at the chosen rate	$15 + 3$	3.3
Step 2: runs with consecutive checkpoints (3 optimizers $\times$ 3 data orders)	9	1.9
Step 3: swaps again, with checkpoints at steps 9998–10,000	18	2.1
Total	141	33.0